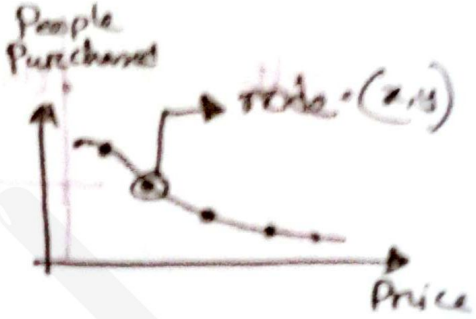
Chapter - 2
Polynomial



Polynomial linear,

$$a + bx + cx^2 + dx^3 + \cdots$$

$f(x) = \sin(x)$; taylor series.

$$f(a) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \frac{f'''(a)}{3!} + \cdots + \frac{f^n(a)}{n!}(x-a)^n + \cdots$$

let, $f(x) = x^2 + 2x + 3 = P_2(x)$
↳ Degree.

It can be represented to infinite dimensions.

$$f(x) \in V^\infty \longrightarrow P_{n+1}(x) \in V^{n+1}$$

↳ Tranquition Error

degree = n, dimension = n + 1

#

$P_3(x) = 5 + 2x + 9x^2 + 15x^3$

Basic Set : $\{x^0, x^1, x^2, x^3\}$

Coefficient Set : $\{5, 2, 9, 15\}$

$5i + 2j + 3k$

$\{\hat{i}, \hat{j}, \hat{k}\}$

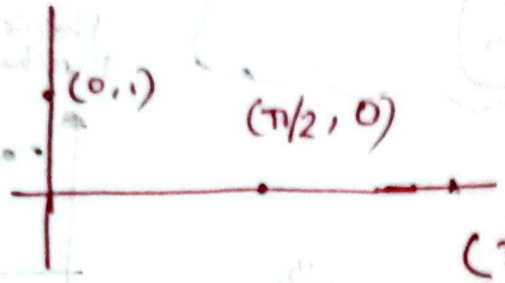
Degree of the Polynomial = Datapoints - 1

= 3 - 1 = 2

$P_3(x) = 5 + 2x + 15x^3$

$\begin{bmatrix} 5 \\ 2 \\ 0 \\ 15 \end{bmatrix}$

#



Here,

Nodes	x	$y = f(x) = P_n(x)$
1	0	1
2	π/2	0
3	π	-1

Here, Dimension = 3 ; Degree = 2

So, $P_2(x) = a_0 + a_1 x + a_2 x^2 = f(x)$

when, $x = 0$.

$a_0 + 0 + 0 = 1 \quad \ldots\ldots (i)$

$x = \pi/2$ $\quad a_0 + \frac{\pi}{2} a_1 + \frac{\pi^2}{4} a_2 = 0 \quad \ldots (ii)$

$x = \pi$ $\quad a_0 + \pi a_1 + \pi^2 a_2 = -1 \quad \ldots (iii)$

So,
$$\begin{bmatrix} 1 & 0 & 0 \\ 1 & \pi/2 & \pi^2/4 \\ 1 & \pi & \pi^2 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & \pi/2 & \pi^2/4 \\ 1 & \pi & \pi^2 \end{bmatrix}^{-1} \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

=

$P_3(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3$

Here, degree = 3 ; coefficient = 4 (3+1)

{dimension = coefficient}

Vector Space

Region where we can → add vectors.
→ multiply with scalers

Basis

is a set of vectors that spans the space.

let, $P_3(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3$

basis = $\{1, x, x^2, x^3\}$ → $2 + 3x + 4x^2 + 5x^3$
↑ row data → $1 + 2x + 9x^2 + 8x^3$

dimensional space = 4

$P_5(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3$

basis = $\{1, x, x^2, x^3, x^4, x^5\}$

dimensional space = 6

Functional space

$f(x) = 2 + 3x + 10x^2 + 14x^3 + 25x^4 + \dots$

$P_n(x) = a_0 + a_1x + a_2x^2 + \dots + a_nx^n$ → as basis: $\{1, x, x^2 \dots\}$

$f(x) \in V^{\infty}$ → [Vector]

So, $P_n(x) \in V^{(n+1)}$ → dimension

let,

$f(x) = 2 + 3x + 4x^2 + 5x^3 + 8x^4 + 9x^5 + \dots$ → Error Part

$P_2(x) = a_0 + a_1x + a_2x^2$

$P_5(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + a_5x^5 \dots$

→ Error Part is degreasing

Degree বাড়লে Error Part কমে Curve

↳ Weierstrass Approx. Theorem

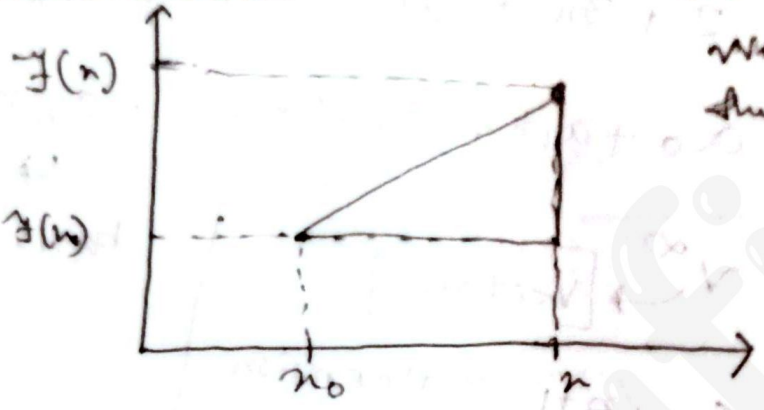
$f(x) \to P_n(x)$	$P_n(x)$ ∝ error ↓

Taylor Series

For Straight line

we know the value of $f(x_0)$, x_0, x, $f'(x_0)$

we need to determine the value of $f(x)$



the slope, gradient $= \frac{y_2 - y_1}{x_2 - x_1}$

$$= \frac{f(x) - f(x_0)}{x - x_0}$$

$$= f'(x_0)$$

$$f(x) = f'(x_0)(x - x_0) + f(x_0)$$

Here, $f(x) = f'(x_0)(x - x_0) + f(x_0)$

#

$$f(x) = f(x_0) + f'(x_0)(x-x_0) + \frac{f''(x_0)(x-x_0)^2}{2!} + \frac{f'''(x_0)(x-x_0)^3}{3!} + \cdots$$

* Proof of Taylor Series

$$f(x) = a_0 + a_1(x-x_0) + a_2(x-x_0)^2 + a_3(x-x_0)^3 + \cdots$$

differentiated val

$$f'(x) = 0 + a_1 + 2a_2(x-x_0) + 3a_3(x-x_3)^2 + \cdots$$

$$f''(x) = 2a_2 + 3\times 2a_3(x-x_0)$$

$$f'''(x) = 3.2\,a_3$$

let, $x = x_0$. so,

$f(x_0) = a_0$

$f'(x_0) = a_1$

$f''(x_0) = 2a_2$

$f'''(x_0) = 3.2.a_3$

So, $a_2 = \frac{f''(x_0)}{2!}$

$a_3 = \frac{f'''(x_0)}{3.2}$

$= \frac{f'''(x_0)}{3!}$

Practice fm → e^x, e^{-x}

Replacing the values of, a_0, a_1, a_2, a_3

we get, $f(x) = f(x_0) + f'(x_0)(x-x_0) + \frac{f''(x_0)(x-x_0)^2}{2!} + \ldots$

$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} + \ldots$

$f(x) = f(x_0) + f'(x_0)(x-x_0) + f''\left(\frac{x_0}{2!}\right)(x-x_0)^2 + f^3\frac{(x_0)}{3!}(x-x_0)^3$

$x_0 = 0$

$\sin 0 + \cos 0\,(x) - \sin 0 - \frac{\cos(0)x^3}{3!}$

We can predict values using polynomial

degree = n, Coefficient = n+1

Chapter - 02

Vandermonde Matrix

$$P_n(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3 \cdots a_n x^n$$

Let

Nodes →

(x) House Sign	Rent (f(x))
20	200
30	230
40	275

Nodes

$f(x) = P_n(x)$

Here, $P_n(30) = 230$

(x) Age	Salary (f(x))
25	25K
30	30K
45	70K

If we are to ask what will be the salary of a man aged 35. We can ~~assum~~ predict predict a value. So, we need to set a polynomial

$$P_n(x) = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n$$

where, $P_n(\text{nodes}) = f(\text{nodes})$

So,

$$P_2(x) = a_0 + a_1 x + a_2 x^2$$

$P_2(25) \Rightarrow a_0 + 25 a_1 + 625 a_2 = 25000$

$P_2(30) \Rightarrow a_0 + 30 a_1 + 900 a_2 = 30000$

$P_2(45) \Rightarrow a_0 + 45 a_1 + 2025 a_2 = 70000$

For (n+1) nodes ; the Polynomial degree will be (n)

or for (n) nodes ; the polynomial degree will be (n-1)

Node অনুযায়ী Degree সর্বসাকুল্যে যত বেশি 1 কমবে।

Time (t)	Velocity f(t)
0	0
10	227.4
15	362.8
20	517.35

Here node = 4 ; so degree = (4-1) = 3

So,

$P_3 = a_0 + a_1 x + a_2 x^2 + a_3 x^3$

↳ This can be solved using vandermonde Matrix

Vandermonde Matrix Issue → We can't solve (It becomes too hard) for nodes greater than 2.

Mainly Solve for P_1

→ Time complexity

Space complexity (যত বেশি)

we

let,

x_0	$f(x_0)$
2	5
3	6

Here, node = 2 ; degree = 1 ; So,

$P_1(x_1) = a_0 + a_1 x_1 = f(x_1)$

$P_1(x_2) = a_0 + a_1 x_2 = f(x_2)$

Now, taking it to the matrix form.

$$\begin{bmatrix} 1 & x_1 \\ 1 & x_2 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} f(x_1) \\ f(x_2) \end{bmatrix}$$

So,

$$\begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \end{bmatrix}^{-1} \begin{bmatrix} f(x_1) \\ f(x_2) \end{bmatrix}$$

[determinant 0 হলে ⇒ matrix কে solve করা যাবে না]

Example

Time	Velocity
15	362.8
20	517.35

Node = 2 ; degree = 1

So,

$P_1(15) \Rightarrow a_0 + 15a_1 = 362.8$

$P_1(20) \Rightarrow a_0 + 20a_1 = 517.35$

Now,

$$\begin{bmatrix} 1 & 15 \\ 1 & 20 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} 362.8 \\ 517.35 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} 1 & 15 \\ 1 & 20 \end{bmatrix}^{-1} \begin{bmatrix} 362.8 \\ 517.35 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \frac{1}{(1\times20)-(1\times15)} \begin{bmatrix} 20 & -15 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 362.8 \\ 517.35 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \frac{1}{5} \begin{bmatrix} 20 & -15 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 362.8 \\ 517.35 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} 4 & -3 \\ -1/5 & 1/5 \end{bmatrix} \begin{bmatrix} 362.8 \\ 517.35 \end{bmatrix}$$

$$\text{So, } \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} -100.85 \\ 30.91 \end{bmatrix}$$

Lagrange To solve the issue for Vandermonde we use this.

	x	$f(x)$
x_0	2	30
x_1	5	45
x_2	7	25

Here, node = 3; degree = 2

$P_2(x) = a_0 + a_1 x + a_2 x^2$

but in lagrange we write it like this ↓

$$P_2(x) = l_0(x) f(x_0) + l_1(x) f(x_1) + l_2(x) f(x_2)$$

Now,

$$l_0(x) = \frac{x - x_1}{x_0 - x_1} \times \frac{x - x_2}{x_0 - x_2}$$

$$l_1(x) = \frac{x - x_0}{x_1 - x_0} \times \frac{x - x_2}{x_1 - x_2}$$

$$l_2(x) = \frac{x - x_0}{x_2 - x_0} \times \frac{x - x_1}{x_2 - x_1}$$

#	Time	Velocity (m/s)
x_0	15	227.04
x_1	20	362.78
x_2	22.5	517.35

So, $P_n(17) = ?$

Here, # nodes = 3 so degree = 2

So,
$$P_2(x) = l_0(x) f(x_0) + l_1(x) f(x_1) + l_2(x) f(x_2)$$

So,
$$l_0(x) = \left(\frac{x-x_1}{x_0-x_1} \times \frac{x-x_2}{x_0-x_2}\right) = \frac{x-20}{15-20} \times \frac{x-22.5}{15-22.5}$$

$$= \frac{2}{75}(x-20)(x-22.5)$$

Now,
$$l_1(x) = \left(\frac{x-x_0}{x_1-x_0}\right) \times \left(\frac{x-x_2}{x_1-x_2}\right) = \frac{x-15}{20-15} \times \frac{x-22.5}{20-22.5}$$

$$= -\frac{2}{25}(x-15)(x-22.5)$$

$$l_2(x) = \left(\frac{x-x_0}{x_2-x_0}\right) \times \left(\frac{x-x_1}{x_2-x_1}\right) = \frac{x-15}{22.5-15} \times \frac{x-20}{22.5-20}$$

$$= \frac{4}{75}(x-15)(x-20)$$

So, $P_2(x) = \frac{2}{75}(x-20)(x-22.5)(227.04) - \frac{2}{25}(x-15)(x-22.5)(362.78) + \frac{4}{75}(x-15)(x-20)(517.35)$ (Ans.)

Here,

$$P_2(x) = \frac{2}{75}(x-20)(x-22.5)(227.04) - \frac{2}{25}(x-15)(x-22.5)(362.74) + \frac{4}{75}(x-15)(x-20)(517.35)$$

$$\text{So, } P_2(17) = \frac{2\times 4}{75}(17-20)(17-22.5)(227.04) - \frac{2\times 362.74}{25}(17-15)(17-22.5) + \frac{4\times 517.35}{75}(x-15)(x-20)$$

$$= 253.592 \quad (\text{Ans.})$$

$f(x) = \cos(x)$; $x_0 = -\pi/4$, $x_1 = 0$, $x_2 = \frac{\pi}{4}$

So, $P_n(x) = ?$

Here, node = 3, So, ~~degree~~ degree = 2

$$P_2(x) = l_0(x) f(x_0) + l_1(x) f(x_1) + l_2(x) f(x_2)$$

Now,

$$l_0(x) = \frac{x-x_1}{x_0-x_1} \times \frac{x-x_2}{x_0-x_2} = \frac{x}{-\pi/4} \times \frac{x-\frac{\pi}{2}}{-\frac{\pi}{4}-\frac{\pi}{4}} = \frac{x}{-\pi/4} \times \frac{x-\frac{\pi}{2}}{-\frac{\pi}{2}}$$

$$= \frac{x}{\pi^2/8}\left(x-\frac{\pi}{2}\right)$$

$$= \frac{8x}{\pi^2}\left(x-\frac{\pi}{2}\right)$$

$(x-0)(x-\frac{\pi}{4}) = x \times (x+\frac{\pi}{4})$

$x+\frac{\pi}{4} = x(x+\frac{\pi}{4})$

$$l_1(x) = \frac{x-x_0}{x_1-x_0} \times \frac{x-x_2}{x_1-x_2} = \frac{x+\pi/4}{0+\pi/4} \times \frac{x-\pi/4}{0-\pi/4}$$

$$= -\frac{16}{\pi^2}\left(x+\frac{\pi}{4}\right)\left(x-\frac{\pi}{4}\right)$$

$$l_2(x) = \frac{x-x_0}{x_2-x_0} \times \frac{x-x_1}{x_2-x_1} = \frac{x+\pi/4}{\pi/4+\pi/4} \times \frac{x-0}{\pi/4-0}$$

$$= \frac{1}{2\pi/4 \times \pi/4}\left(x+\pi/4\right)$$

$$= \frac{8}{\pi^2}\left(x+\frac{\pi}{4}\right)$$

Here, $x_0 = -\pi/4$; $f(x_0) = \cos(-\pi/4) = \frac{1}{\sqrt{2}}$

$x_1 = 0$; $f(x_1) = \cos(0) = 1$

$x_2 = \pi/4$; $f(x_2) = \cos(\pi/4) = \frac{1}{\sqrt{2}}$

So, $P_2(x) = \frac{8x}{\pi^2}\left(x-\frac{\pi}{2}\right)\left(\frac{1}{\sqrt{2}}\right) - \frac{16}{\pi^2}\left(x+\frac{\pi}{4}\right)\left(x-\frac{\pi}{4}\right)(1)$

$$+\frac{8x}{\pi^2}\left(x+\frac{\pi}{4}\right)\left(\frac{1}{\sqrt{2}}\right)$$

If	Time	Velocity
16	0	0
6	10	227.4
1	15	362.8
4	20	517.35
6.5	22.5	602.97
14	30	901.67

If asks $P_2(x)$ then calculate 1st 3 (0, 10, 15); if it asks find $P_2(16)$ or $P_2(x)$ at $x=16$, then find the nearest (10, 15, 20)

Here, $P_2(x) = l_0(x) f(x_0) + l_1(x) f(x_1) + l_2(x) f(x_2)$

Here, $x_0 = 10$, $x_1 = 15$, $x_2 = 20$

So,

$$l_0(x) = \frac{x-x_1}{x_0-x_1} \times \frac{x-x_2}{x_0-x_2} = \frac{x-15}{10-15} \times \frac{x-20}{10-20}$$

$$= \frac{x-15}{-5} \times \frac{x-20}{-10}$$

$$= \frac{(x-15)(x-20)}{50}$$

$$= \frac{(x-5)(x-4)}{10}$$

$x_0 = 10$

$x_1 = 15$ $\quad x_2 = 20$

So, $l_1(x) = \frac{x-x_0}{x_1-x_0} \times \frac{x-x_2}{x_1-x_2} = \frac{x-10}{15-10} \times \frac{x-20}{15-20}$

$= \frac{(x-10)(x-20)}{5 \times (-5)}$ =

$= \frac{(x-10)(x-20)}{-25}$

$l_2(x) = \frac{x-x_0}{x_2-x_0} \times \frac{x-x_1}{x_2-x_1} = \frac{x-10}{20-10} \times \frac{x-15}{20-15}$

$= \frac{(x-10)(x-15)}{50}$

So, $P_2(x) = \frac{(x-15)(x-20)}{50} f(x_0) - \frac{(x-10)(x-20)}{25} f(x_1)$

$+ \frac{(x-10)(x-15)}{50} f(x_2)$

So, $P_2(16) = \frac{(16-15)(16-20)}{50}(227.4) - \frac{(16-10)(16-20)}{25}(362.8)$

$+ \frac{(16-10)(16-20)}{50}(517.35)$

$= \frac{14763}{125}$; 118.152

Kronekor Delta

Kronekor Delta

x_0	$f(x)$
x_1	$f(x_1)$

So, $P_1(x) = l_0(x) f(x_0) + l_1(x) f(x_1)$

So, $l_0(x) = \frac{x - x_1}{x_0 - x_1}$ $\left\{ \begin{matrix} l_0(x_0) = 1 \\ l_0(x_1) = 0 \end{matrix} \right\}$ $l_0(x_i) \begin{cases} 0 ; i = 1 \\ 1 ; i = 0 \end{cases}$

$l_1(x) = \frac{x - x_0}{x_1 - x_0}$ $\left\{ \begin{matrix} l_1(x_0) = 0 \\ l_1(x_1) = 1 \end{matrix} \right\}$ $l_1(x_i) \begin{cases} 0 ; i = 0 \\ 1 ; i = 1 \end{cases}$

So, $l_n(x_j) = \delta_{nj}$

Now, $\delta_{nj} = 0 \; ; \; n \neq j$

$\delta_{nj} = 1 \; ; \; n = j$

Newton's Divided difference & Cauchy's Theorem

Time Complexity $O(n^2)$

Here, we write the polynomial as →

$$P_n(x) = a_0 + a_1(x-x_0) + a_2(x-x_0)(x-x_1) + a_3(x-x_0)(x-x_1)(x-x_2) + \cdots + a_n(x-x_0)(x-x_1)\cdots(x-x_n)$$

Here,

$a_0 = f[x_0]$

$a_1 = f[x_0, x_1]$

$a_2 = f[x_0, x_1, x_2]$

$a_3 = f[x_0, x_1, x_2, x_3]$

So, $a_n = f[x_0, x_1, \ldots, x_n]$

Right way to write this →

$$P_4(x) = f[x_0] + f[x_0, x_1](x-x_0) + f[x_0, x_1, x_2](x-x_0)(x-x_1) + f[x_0, x_1, x_2, x_3](x-x_0)(x-x_1)(x-x_2) + f[x_0, x_1, x_2, x_3, x_4](x-x_0)(x-x_1)(x-x_2)(x-x_3)$$

#

	x	$f(x)$
$x_0 \rightarrow$	-1	5
$x_1 \rightarrow$	0	1
$x_2 \rightarrow$	1	3
$x_3 \rightarrow$	2	11

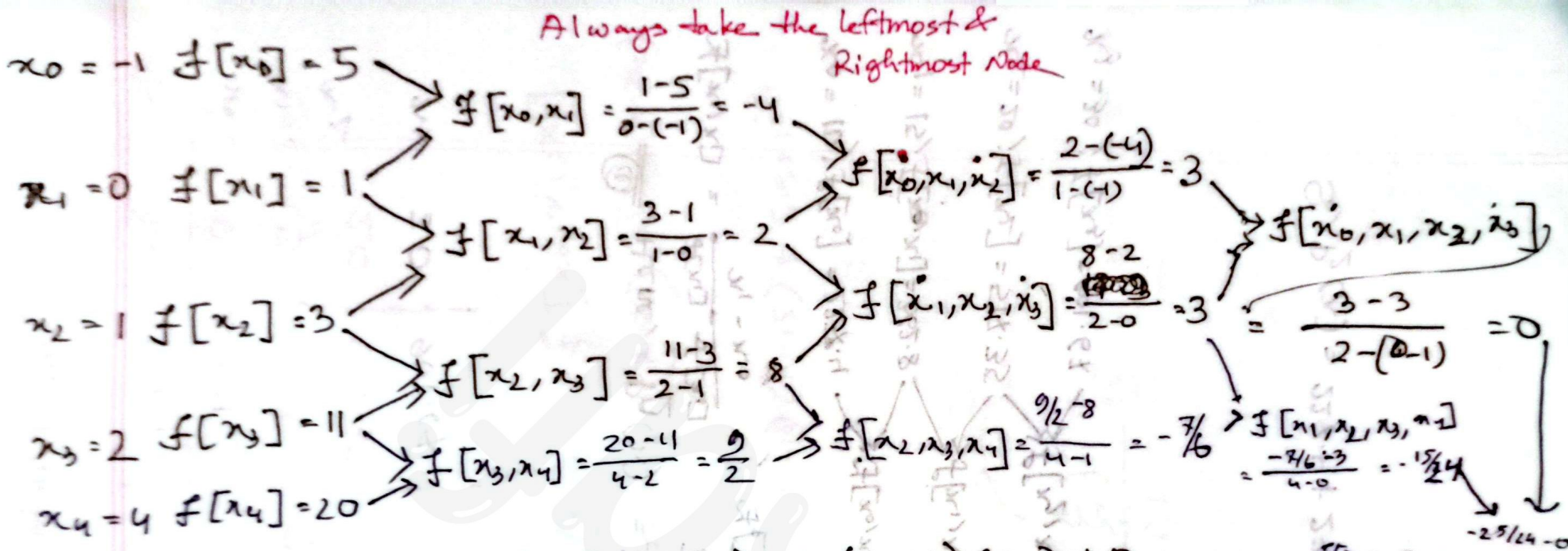
So, $P_3(x) = f[x_0] + f[x_0, x_1](x-x_0) + f[x_0, x_1, x_2](x-x_0)(x-x_1) + f[x_0, x_1, x_2, x_3](x-x_0)(x-x_1)(x-x_2)$

So, $P_3(x) = f[x_0] + f[x_0, x_1](x-x_0) + f[x_0, x_1, x_2](x-x_0)(x-x_1) + f[x_0, x_1, x_2](x-x_0)(x-x_1)(x-x_2)$

$+ f[x_0, x_1, x_2, x_3](x-x_0)(x-x_1)(x-x_2)(x-x_3)$

The calculation done in the next page →

Always take the leftmost & Rightmost Node



$$S_0, P_3(x) = 5 + (-4)(x+1) + 3(x+1)(x) + 0 + \left(-\frac{5}{24}\right)(x+1)(x)(x-1)(x-2)$$

(Ans.)

আরেকটি কথা আমাদের New node add করার সময় বলে দিয়েছিলাম $x_4 \to 4$; 20. Lagrange এ আমরা সেটা করি নাই calculate করা লাগছে। But in here, just add in the list

Using Newton's formula

#	Time	Velocity
x_0	10	227.4
x_1	15	362.8
x_2	20	517.35
x_3	30	901.67

ⓐ Find the Polynomial expression.
$P_n(x) = ?$

ⓑ $P_2(17) = ?$

ⓒ Add a new datapoint (30, 901.67)

ⓐ Here, $P_2(x) = f[x_0] + f[x_0, x_1](x - x_0) + f[x_0, x_1, x_2](x - x_0)(x - x_1)$

$f[x_0, x_1] = \frac{f[x_1] - f[x_0]}{x_1 - x_0}$; $f[x_0, x_1, x_2] = \frac{f[x_2] - f[x_0]}{x_2 - x_0}$

$x_0 = 10$; $f[x_0] = 227.4$
$x_1 = 15$; $f[x_1] = 362.8$
$x_2 = 20$; $f[x_2] = 517.35$
$x_3 = 30$; $f[x_3] = 901.67$

$f[x_0, x_1] = 27.08$
$f[x_1, x_2] = 30.91$
$f[x_2, x_3] = 38.432$

$f[x_0, x_1, x_2] = 0.383$
$f[x_1, x_2, x_3] = 0.501$

$f[x_0, x_1, x_2, x_3] = \frac{0.501 - 0.383}{30 - 10}$

$= 5.9 \times 10^{-3}$

So, $P_2(x) = 227.4 + 27.08(x - 10) + 0.383(x - 10)(x - 15)$

(b)

$$P_2(x) = 227.4 + 27.08(x-10) + 0.383(x-10)(x-15)$$

So, $P_2(17) = 227.4 + 27.08(17-10) + 0.383(17-10)(17-15)$

$$= 422.322$$

(c) So,

$$P_3(x) = f[x_0] + f[x_0, x_1](x-x_0) + f[x_0, x_1, x_2](x-x_0)(x-x_1) + f[x_0, x_1, x_2](x-x_0)(x-x_1)(x-x_2)$$

$$= 227.4 + 27.08(x-10) + 0.383(x-10)(x-15) + 5.5\times10^{-3}(x-10)(x-15)(x-20)$$

Cauchy Theorem → Error

$$|f(x) - P_n(x)| = \frac{f^{n+1}(\xi)}{(n+1)!}(x-x_0)(x-x_1)\cdots(x-x_n)$$

(→ MAX)

$|f(x) - P_n(x)|$ → infinite

$x_0 \quad x_1 \quad x_2$

Ex $f(x) = \cos x$; $\{-\frac{\pi}{4}, 0, \frac{\pi}{4}\}$ Calculate the upper bound error using cauchys theorem.

[Range] $x/\xi \in [-1, 1]$

Here,

$$|f(x) - P_2(x)| = \frac{f^3(\xi)}{3!}(x-x_0)(x-x_1)(x-x_2)$$

$$= \left|\frac{\sin(\xi)}{3!}\right| \left|(x+\tfrac{\pi}{4})(x)(x-\tfrac{\pi}{4})\right|$$

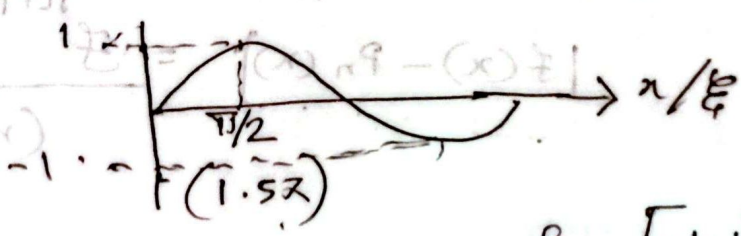
Now, $f(x) = \cos x$

$\Rightarrow f'(x) = -\sin x$

$\Rightarrow f''(x) = -\cos x$

$\Rightarrow f'''(x) = \sin x$

Now, $\frac{\sin(\xi)}{3!} = 1.57$



$= \frac{1}{3!}$

but as it's not in the range of $[-1, 1]$

So, $\frac{\sin(1)}{3!}$

$w(x) = (x - \frac{\pi}{4})\cdot(x - 0)\,(x - \frac{\pi}{4})$

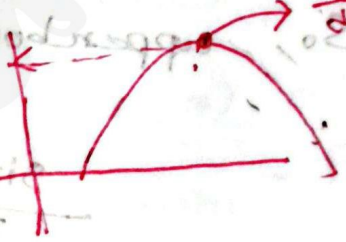
Now,

$$w(x) = \left(x + \frac{\pi}{4}\right)(x)\left(x - \frac{\pi}{4}\right) = (x)\left(x^2 - \frac{\pi^2}{16}\right)$$

$$= x^3 - \pi^2/16\,(x)$$

So, $w(x) = x^3 - \frac{\pi^2}{16}x$

to get the maximum point of a function, the derivative has to be 0. ; as $y = x^2 + 3x + 1$ should be 0 after defenentiate



So, $\frac{dy}{dx} = 0$ হলে, x এর value solve করলে, যাকে ঐ x এর value এ function এর max পাব.

So, $w(x) = x^3 - \frac{\pi^2}{16}x$

$$\Rightarrow w'(x) = 3x^2 - \frac{\pi^2}{16} = 0$$

$$\text{So, } x = \pm \frac{\pi}{4\sqrt{3}}$$

x	$\lvert w(x) \rvert$
$-\pi/4\sqrt{3}$	0.186
$\pi/4\sqrt{3}$	0.186
-1	0.383
1	0.383

Highest value

So,

$$|f(x) - P_2(x)| = \frac{f^3(\xi)}{3!}(x-x_0)(x-x_1)(x-x_2)$$

$$= \left|\frac{\sin(\xi)}{3!}\right| \left|\left(x+\frac{\pi}{4}\right)(x)\left(x-\frac{\pi}{4}\right)\right|$$

$$= \left|\frac{\sin(1)}{3!}\right| \times 0.383$$

So, upperbound of the error,

$$\frac{\sin(1)}{3!} \times 0.383$$

The formula

$$|f_n(x) - P_n(x)| = \left|\frac{f^{n+1}(\xi)}{(n+1)!}\right| (x-x_0)(x-x_1)\cdots(x-x_n)$$

Hermite Interpolation

x	$f(x)$	$f'(x)$
3	15	40
5	27	20
9	35	37

Here are (3+3) conditions,

So, Polynomial will be, $P_{6-1} = P_5$

Here, $P_{2n+1} = P_{2(2)+1} = P_5$

	$f(x)$	$f'(x)$
x_0	$f(x_0)$	$f'(x_0)$
x_1	$f(x_1)$	$f'(x_1)$
x_2	$f(x_2)$	$f'(x_2)$
x_3	$f(x_3)$	$f'(x_3)$

So, $(n = 3)$

$P_{2n+1} = P_{2(3)+1} = P_7$

Ex $f(x) = \sin(x)$

So, $P_3(x) = ?$

Hermite interpolation

	x	$f(x)$	$f'(x)$
x_0	0	0	1
x_1	$\pi/2$	1	0

~~$P_3(x) = h_0(x) f(x_0) + h_1(x) f(x_1)$~~

$P_3(x) = h_0(x) f(x_0) + h_1 f(x_1) + \hat{h}_0(x) f'(x_0) + \hat{h}_1 f'(x_1)$ (→0)

So, $P_3(x) = h_1(x) f(x_1) + \hat{h}_0(x) f'(x_0)$

Formula

$$h_k(x) = \{1 - 2(x - x_k) l_k'(x_k)\} (l_k(x))^2$$

$$\hat{h}_k(x) = (x - x_k) (l_k(x))^2$$

Now,

$h_1(x) \Rightarrow$

$$l_1(x) = \frac{x - x_0}{x_1 - x_0} = \frac{x-0}{\frac{\pi}{2} - 0} = \frac{x}{\pi/2} = \frac{2}{\pi} x$$

$$l_1'(x) = \frac{2}{\pi} \quad ; x_1 = \frac{\pi}{2}$$

$$\text{So, } h_1(x) = \left[1 - 2\left(x - \frac{\pi}{2}\right)\left(\frac{2}{\pi}\right)\right] \left(\frac{2}{\pi} x\right)^2$$

$\hat{h}_0(x) \Rightarrow$

$$\hat{h}_k(x) = (x - x_k)\left(l_k(x)\right)^2$$

$$\hat{h}_0(x) =$$

$$l_0(x) = \frac{x - x_1}{x_0 - x_1} = \frac{x - \frac{\pi}{2}}{0 - \frac{\pi}{2}} = 1 - \frac{2}{\pi} x$$

So,

$$\hat{h}_0(x) = (x - 0)\left(1 - \frac{2}{\pi} x\right)^2$$

$$\text{So, } P_3(x) = h_1(x) f(x_1) + \hat{h}_0(x) f'(x_0)$$

$$= \left\{1 - \frac{y}{\pi}\left(x - \frac{\pi}{2}\right)\right\}\left(\frac{2}{\pi} - x\right)^2 (1)$$

$$+ \quad x\left(1 - \frac{2}{\pi} x\right)^2 \cdot (1)$$

Ex

	x	$f(x)$	$f'(x)$
x_0	-1	1	2
x_1	0	0	2
x_2	1	1	0

P_5

We know,

$$P_5(x) = h_0(x) f(x_0) + h_1(x) f(x_1) + h_2(x) f(x_2) + \hat{h}_0(x) f'(x_0) + \hat{h}_1(x) f'(x_1) + \hat{h}_2(x) f'(x_2)$$

($h_1(x) f(x_1)$ and $\hat{h}_2(x) f'(x_2)$ vanish)

So,

$$P_5(x) = h_0(x) f(x_0) + h_2(x) f(x_2) + \hat{h}_0(x) f'(x_0) + \hat{h}_1(x) f'(x_1)$$

We know,

$$h_k(x) = \{1 - 2(x - x_k) l_k'(x_k)\} (l_k(x))^2$$

$$\hat{h}_k(x) = (x - x_k)(l_k(x))^2$$

Now

$l_0(x)$

$$l_0(x) = \frac{x-x_1}{x_0-x_1} \times \frac{x-x_2}{x_0-x_2} = \frac{x-(0)}{-1-0} \times \frac{x-1}{-1-1} = \frac{1}{2}x(x-1)$$

$$= \frac{1}{2}x^2 - \frac{1}{2}x$$

$$l_0(x) = \frac{1}{2}x^2 - \frac{1}{2}x$$

$$\text{So, } l_0'(x) = x - \frac{1}{2} \quad ; \quad l_0'(x_0) = l_0'(-1) = (-1) - \frac{1}{2} = -\frac{3}{2}$$

$$\text{So, } h_0(x) = \{1 - 2(x-x_0)\, l_0'(x_0)\}\,(l_0(x))^2$$

$$= \{1 - 2(x-(-1))\left(-\frac{3}{2}\right)\}\left(\frac{1}{2}x^2 - \frac{1}{2}x\right)^2$$

$$= 1 + 3(x+1)\left(\frac{1}{2}x^2 - \frac{1}{2}x\right)^2$$

$h_2(x)$

$$l_2(x) = \frac{x-x_0}{x_2-x_0} \times \frac{x-x_1}{x_2-x_1} = \frac{x-(-1)}{1-(-1)} \times \frac{x-0}{1-0} = \frac{x+1}{2} \cdot \frac{x}{1}$$

$$l_2(x) = \frac{1}{2}x(x+1)$$

$$l_2(x) = \frac{1}{2}x^2 + \frac{1}{2}x$$

$$l_2'(x) = x + \frac{1}{2} \quad ; \quad l_2'(x_2) = l_2'(1) = \frac{3}{2}$$

$$\text{So, } h_2(x) = \{1 - 2(x-x_2)\, l_2'(x_2)\}\,(l_2(x))^2$$

$$= \{1 - 2(x-1)\frac{3}{2}\}\left(\frac{1}{2}x^2 + \frac{1}{2}x\right)^2$$

$$= 1 - 3(x-1)\cdot\left(\frac{1}{2}x^2 + \frac{1}{2}x\right)^2$$

$\hat{h}_0(x)$

$$\hat{h}_k(x) = (x - x_k)\,(l_k(x))^2$$

So, $l_0 x = \dfrac{x - x_1}{x_0 - x_1} \times \dfrac{x - x_2}{x_0 - x_2} = \dfrac{x+0}{-1+0} \times \dfrac{x-1}{-1-1}$

$$= \frac{1}{2}\, x\,(x-1) = \frac{1}{2}x^2 - \frac{1}{2}x$$

So, $\hat{h}_0(x) = (x - x_0)\left(\frac{1}{2}x^2 - \frac{1}{2}x\right)^2$

$$= (x+1)\left(\frac{1}{2}x^2 - \frac{1}{2}x\right)^2$$

$l_1 = \dfrac{x - x_0}{x_1 - x_0} \times \dfrac{x - x_2}{x_1 - x_2}$

$= \dfrac{x+1}{0+1} \times \dfrac{x-1}{0-1}$

$= (x+1)(1-x)$

$= 1 - x^2$

$\hat{h}_1(x)$

$$\hat{h}_k(x) = (x - x_k)\left(l_k(x)\right)^2$$

So, $\hat{h}_1(x) = (x - x_1)\,(l_1(x))^2$

$$= (x - 0)\,(1 - x)^2$$

$$= x\,(1-x)^2$$

Runge function →

$$f(x) = \frac{3}{4+3x^2}$$

Interval ↓ $[-4, 4]$

$n = 3$

(a) Calculate the equal angled points / thetas.

(b) Calculate the value of chebyshev nodes.

(c) Find the lagrange baris, $l_2(x)$

Soln

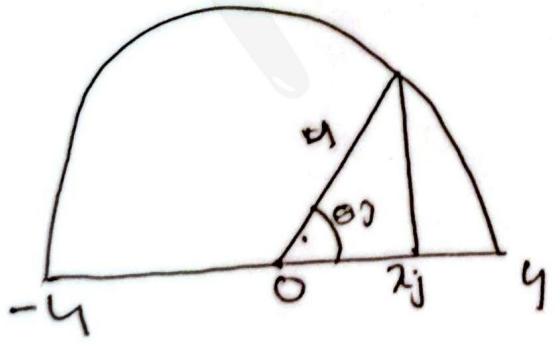
(a)

$$\theta_j = \frac{(2j+1)\pi}{2(n+1)} \quad ; \quad j = 0, 1, 2, 3$$

So, $\theta_0 = \frac{1}{8}\pi$; $\theta_1 = \frac{3}{8}\pi$

$\theta_2 = \frac{5}{8}\pi$; $\theta_4 = \frac{7}{8}\pi$

(b)



$$\cos\theta_j = \frac{x_j}{4}$$

$$\therefore x_j = 4\cos\theta_j$$

$$\therefore x_0 = 4\cos\left(\frac{1}{8}\pi\right) \quad ; \quad x_1 = 4\cos\left(\frac{3}{8}\pi\right)$$

$$x_2 = 4\cos\left(\frac{5}{8}\pi\right) \quad ; \quad x_4 = 4\cos\left(\frac{7}{8}\pi\right)$$

(c)

Here,

	x	$f(x)$
$x_0 \longrightarrow$	$4\cos\left(\frac{1}{8}\pi\right)$	
$x_1 \longrightarrow$	$4\cos\left(\frac{3}{8}\pi\right)$	
$x_2 \longrightarrow$	$4\cos\left(\frac{5}{8}\pi\right)$	
$x_3 \longrightarrow$	$4\cos\left(\frac{7}{8}\pi\right)$	

$$\therefore \quad l_2(x) = \left(\frac{x - x_0}{x_2 - x_0}\right) \times \left(\frac{x - x_1}{x_2 - x_1}\right) \times \left(\frac{x - x_3}{x_2 - x_3}\right)$$